AWS Assignment

Course Name & Code: AWS

Instructor: Nitha

Assignment Details

Assignment Title: Task #48804 - Booting process ,Ports and Server Admin commands

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📝 Instructions and Guidelines

Objective

- Booting Process in Linux (Steps in detail)

- Important Port Numbers

- Some Advanced Server Administrator commands for troubleshooting and monitoring

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Result

Step 1: BIOS / UEFI

- BIOS (Basic Input/Output System): Performs a POST (Power-On Self-Test) to check hardware like RAM and CPU. It then looks for a bootable device (Hard Drive, USB) based on the "Boot Priority."
- UEFI (Unified Extensible Firmware Interface): The modern successor to BIOS. It is faster, supports larger disks (GPT), and can load bootloaders directly without needing a Master Boot Record (MBR).

Step 2: Bootloader (GRUB2)

- Once hardwares are initialized, the BIOS/UEFI loads the GRUB2 (Grand Unified Bootloader) into memory.
- Role: GRUB allows you to select which OS or kernel version to boot. It then loads the selected Kernel and the initrd (Initial RAM Disk) into memory.

Step 3: Kernel Execution

- The Kernel initializes hardware drivers and mounts the root filesystem (/) in read-only mode initially.
- It uses the initrd/initramfs to load necessary drivers that aren't built into the kernel yet.

Step 4: Init (The systemd Stage)

- The kernel starts the first process, known as init. In almost all modern Linux distributions this is systemd.
- PID 1: systemd always has a Process ID of 1. It is responsible for bringing the system to a usable state by starting services (daemons), mounting filesystems, and managing the network.

Step 5: Runlevels / Targets

- systemd uses Targets (equivalent to old Runlevels) to determine what state the system should be in:
    - multi-user.target: Standard command-line mode.
    - graphical.target: GUI mode (Desktop environment).

3. Important Port Numbers

Port	Protocol	Description
20/21	FTP	File Transfer Protocol (Data/Command)
22	SSH	Secure Shell for remote access
25	SMTP	Simple Mail Transfer Protocol
53	DNS	Domain Name System
80	HTTP	Hypertext Transfer Protocol (Unsecured Web)
110	POP3	Post Office Protocol (Email)
143	IMAP	Internet Message Access Protocol (Email)
443	HTTPS	Hypertext Transfer Protocol Secure (Encrypted Web)
3306	MySQL	Default port for MySQL/MariaDB databases
5432	PostgreSQL	Default port for PostgreSQL databases

4. 20 Essential Linux Monitoring Commands

These tools are used to troubleshoot performance and health.

Commands	**Use**
top	Real-time system monitoring.
htop	Interactive, user-friendly process viewer.
uptime	Displays system load and run time.
free -m	Checks RAM and Swap usage.
df -h	Monitors disk space usage.
du -sh	Estimates directory space usage.
iostat	Reports CPU and I/O statistics.
iotop	Monitors disk I/O per process.
vmstat	Displays virtual memory statistics.
mpstat	Shows CPU usage per individual core.
netstat -tulnp	Lists active network connections and ports.
ss	A faster alternative to netstat.
ip addr	Displays network interface details.
ps aux	Captures a snapshot of all running processes.
lsof	Lists all open files and network sockets.
nload	Visualizes real-time network traffic.
sar	Collects and reports system activity history.
lsblk	Lists block devices (hard drives and partitions).
dmesg	Views kernel ring buffer messages for hardware errors.
journalctl -f	Real-time tracking of system logs via systemd.